

Combining Natural Techniques to Achieve Seamless Locomotion in Consumer VR Spaces

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ABSTRACT

We introduce a seamless locomotion concept designed for a commercial VR game. The locomotion concept was designed with the following goal: The whole game can be explored entirely without teleportation or joystick steering but only by natural locomotion. This should create a seamless fusion of exploration and discovery and, therefore, maximize presence and accessibility. Therefore, our game leverages a mix of different techniques: 1) *Redirected Walking techniques* such as rotation gains and impossible spaces, 2) *Passive travel* like lifts, moving platforms, vehicles, etc., 3) *Indirect movement* such as climbing, pulling a rope to move a raft, etc. Furthermore, these techniques are extended by or coupled with physical interactions or movements like crawling, jumping or sliding around corners, to distract player during the manipulation. The seamless integration of all these techniques enables the player to walk around freely in the recommended room-scale play space of 2x2 meters, while exploring a potentially infinite virtual space.

Keywords: Virtual Reality, Locomotion, Game Design

Index Terms: Human-centered computing [Interaction Design]: Interaction Design Process and Methods—User Interface Design; Human-centered computing [Human computer interaction (HCI)]: Interaction Paradigms—Virtual Reality;

1 INTRODUCTION

There is a lot of evidence that natural walking is an advantageous locomotion technique facing criteria like presence, VR sickness, wayfinding, spatial cognition, or usability [2]. In addition, it is very easy to learn (most people learn it at the age of one year). However, natural walking in VR is limited by the available physical space surrounding the user. Especially in homes play space is very rare and walking is often restricted to a few square meters.

There are a few techniques that are based on natural walking and try to overcome this problem such as redirected walking (RDW) [4], change blindness illusions [7], or impossible spaces [8]. RDW is a locomotion technique that enables natural walking in confined spaces by manipulating the user's visually perceived movements [4]. For instance, the user's real rotation could be multiplied with a factor to strengthen or weaken the rotation in the virtual environment. Up to certain detection thresholds, such manipulations remain unnoticeable [6]. Unfortunately, these techniques still require large spaces of up to 40x40 meters [6]. Impossible spaces are a technique that is based on scene layouts with overlapping rooms, which would be impossible in the real world. It was found that an overlap of up to 50% is not noticed by most users [8].

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Figure 1: Player is walking around in a confined play space of only 2x2 square meters while exploring a much larger virtual environment by using a mix of different natural locomotion techniques.

While these redirection techniques were mostly developed in research labs, the games industry itself also came up with innovative new techniques that performed quite good for their purpose. One example for this is the climbing metaphor in Crytek's The Climb [1]. A similar approach to impossible spaces is used by the VR game Unseen Diplomacy [3]. It arranges the (self-overlapping) rooms in such a way that the player does not recognize that she is only walking back and forth in a 3m×4m large area. The game Tea for God even creates their impossible spaces procedurally for each run and can handle flexible tracking space sizes [5]. Unfortunately, these games do not pay much attention to make the overlaps subtle.

2 LOCOMOTION APPROACH

We designed a locomotion concept for a commercial VR adventure game and showed how a mix of natural locomotion techniques can lead to a seamless virtual experience. Our approach is to bring together a bunch of different locomotion techniques from research and industry to make natural movements happen in a play space of only 2x2 meters, where each technique alone can not fulfill this demand. We wanted to maximize presence and accessibility. So, we decided that all interactions in the game have to be based on metaphors from the real world. And as far as possible, they have to incorporate the same motor actions as the corresponding action in the real world.

We evaluated techniques from research as well as games and picked those that can be grouped into these three categories:

- **Redirection techniques:** Impossible Spaces and Rotation Gains combined with interactive distractors and integrated into gameplay. For example, in one scene of the game, the player stands in front of a boarded up entrance to a ride. She is able to grab each plank that was mounted in front of this entrance, turn around, and throw it on a garbage heap behind her. Since there are lots of planks, she needs to do this several



Figure 2: Locomotion techniques from left to right: Vehicle, elevator, ladder climbing, horizontal pennant chain climbing.

times before she can enter the ride. During this gameplay task, a rotation gain is active. This means, each time she turns to the right, virtual rotation is slightly faster than real head rotation, and each time she turns to the left, virtual rotation is slightly slower than real head rotation. Hence, the overall rotation is in the same direction and after all planks are removed the player is rotated 180 degrees in the real world but still facing the same direction in the virtual environment. Now, she can continue walking forward which was not possible before cause she was at the border of the play space.

- **Passive movement techniques:** Passive travel techniques are techniques that do not move the player itself but a platform or vehicle which in turn moves the player. These techniques need to be used carefully since they can induce a visual-vestibular conflict in the player: She visually perceives motion but her vestibular system does not register this motion.
- **Indirect movement techniques:** We defined indirect movement as a class of techniques that do not move the player directly but actually the environment. The player selects a point in the world. Then, the world is translated around this point according to the movements of the player. We used this for ladder climbing or to shimmy along a horizontal pennant chain.

We wanted to offer variety, regarding the interactions in the game world, the environmental design, but also regarding the order of the locomotion techniques. It is just more interesting to be confronted with changing tasks and awakens the spirit of discovery. Moreover, it is just necessary to vary locomotion techniques in certain situations. For example, when the player comes out of a tunnel that triggered an impossible space, there would be no space available to directly go into the next overlapping space. So, another technique, e.g. climbing, needs to be used first to bring the player in the right position for the next impossible spaces construct.

Our goal was to create an environment in which the player has a good spatial impression. It should be understandable where the player comes from and where she is supposed to go. This arised amongst other things from games that are already using impossi-

ble spaces, e.g. *Tea for God*, but only offer narrow corridors and claustrophobic labyrinths in which the player loses orientation.

3 CONCLUSION

We did not invent any new technique but combined already known locomotion techniques in a novel and innovative fashion to enable a seamless and natural experience. We paid particular attention to integrating interesting and physical gameplay mechanics within the locomotion.

This resulted in a demo level of more than 30 explorable square meters including impossible spaces, rotation gains, climbing, vehicles, and elevators, while the game requires only 2x2 meters of play space in the real world. However, this is not to be understood as a limit. With our approach, we could just add more locomotion parts to extend the walkable environment as large as we want it to be. The only limitation is that the levels need to be designed according to the used locomotion techniques.

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